

Optimizing employee promotion predictions using machine learning

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Abstract:

Employee promotions are essential for both organizational growth and individual career advancement, yet they often face challenges such as data imbalances and the lack of effective predictive frameworks. This study addresses these issues by applying advanced machine learning models to improve decision-making in human resource management. Using a dataset comprising 54 808 employee records, the study evaluates eight models, including Random Forest, Logistic Regression, SVM, AdaBoost, XGBoost, Gradient Boosting, Decision Tree, and Artificial Neural Networks (ANN). RF and XGBoost emerged as the most effective, with Random Forest achieving an accuracy of 96.21% and XGBoost closely following at 95.96%. Both models demonstrated strong AUC-ROC scores, highlighting their ability to handle complex data patterns. Key features influencing promotion outcomes, such as "Previous year rating" and "Average training score", were identified as critical variables. Advanced balancing techniques such as SMOTE further improved the detection of underrepresented promoted employees, contributing to fairer evaluations. The study's comprehensive framework, which includes detailed feature analysis and mathematical explanations, provides a practical guide for HR systems seeking to optimize promotion processes. Future research could explore hybrid deep learning models like LSTM and CNNs to enhance scalability and predictive power. Additionally, incorporating factors like employee demographics and ethical considerations would foster fairness and transparency in promotion practices, broadening the application of machine learning in HR management.

Key words:

Employee promotion, human resources management, artificial intelligence machine learning, prediction.

1. Introduction

Artificial intelligence (AI) has profoundly transformed Human Resource Management (HRM), reshaping processes such as recruitment, performance evaluations, and career development (Benabou & Touhami, 2025). By processing vast and complex datasets, AI enables organizations to adopt data-driven decision-making strategies that enhance efficiency and transparency, offering tailored solutions that align with their goals (Stone et al., 2024). These innovations have reduced operational costs while optimizing analytics capabilities (Nawaz

et al., 2024). However, challenges such as biases, lack of transparency, and ethical concerns persist, often limiting AI's broader adoption in HRM practices (Rodgers et al., 2023).

Indeed, promotion decisions are central to career progression and workforce planning. As career models evolve toward flexibility and continuous skill development, organizations face pressure to design promotion systems that are both fair and inclusive (Donohue & Tham, 2019). Machine learning (ML) offers practical support by detecting subtle patterns in imbalanced data and by informing equitable, evidence-based decisions and uncovering subtle

patterns often missed by traditional approaches (Barducci et al., 2022). A systematic review identifies Random Forest, Support Vector Machines, and Artificial Neural Networks among the most effective HRM algorithms for their strong predictive capabilities, with applications ranging from turnover to talent acquisition (Nosratabadi et al., 2022; Fallucchi et al., 2020).

Within promotion prediction, Random Forest often performs strongly on large, imbalanced datasets and highlights drivers such as performance ratings and training scores (Chen et al., 2023; Hickey et al., 2022). Gradient Boosting captures nonlinear relationships that matter for accurate promotion estimates (Zhanuzakov & Balakaeva, 2022). Neural Networks can uncover complex patterns, though limited interpretability hinders practical uptake (Şahinbaş, 2022). Recent work reports high accuracy from a Neural Network optimized with the Neural Network Synchronization Optimization Algorithm (NNSOA) (Zhao et al., 2021), and Decision Tree models have shown utility for promotion-related decisions in educational settings (Li, 2024). Despite these advances, comparative evaluations that combine strong predictive performance with class-imbalance control and transparent explanations remain limited for large real-world promotion datasets.

This article examines ML models for employee promotion prediction using a dataset of 54 808 employee records. It makes two contributions. First, it offers a head-to-head evaluation of widely used ML algorithms together with recent enhancements. Second, it incorporates others innovative models and optimization techniques to improve predictive accuracy and address critical challenges, such as class imbalance and model interpretability which will support defensible HR decision-making. Based on the above-mentioned needs and gaps, this study aims to answer the following research questions:

- RQ1. Among the evaluated ML models, which achieve the strongest predictive performance for promotion prediction on this dataset?
- RQ2. Which features emerge as the most influential across models in predicting promotions?

The rest of this study are structured as follows. Section 2 reviews related works and theoretical background. Section 3 introduces the dataset, ML models used and methodology followed in this

study. Section 4 presents the results, including model performance and feature importance. Section 5 provides a detailed discussion, analyzing the implications of the findings. Finally, Section 6 concludes with the key contributions and suggestions for future research.

2. Literature review

2.1. Overview of ML in HRM

Machine Learning models, such as Decision Trees, have demonstrated significant value in HRM by providing transparent and actionable evaluations for promotion decisions. These tools analyze variables including professional experience, hierarchical levels, and performance history, supporting fair and well-informed decision-making processes (Gryncewicz et al., 2023). Advanced techniques like Adaptive Boosting and Random Forest further enhance this capability by identifying critical factors such as seniority and opportunities for professional development, allowing HR professionals to refine promotion strategies that prioritize merit and growth potential (Nosratabadi et al., 2022).

Furthermore, ML methods extend their utility to retention and workforce well-being. Techniques like the Extreme Learning Adaptive Neuro-Fuzzy Inference System (ELANFIS) and Support Vector Machines are applied to identify attrition risks and employee dissatisfaction. These tools equip HR managers with actionable strategies to enhance work conditions, improve retention, and ensure organizational stability (Zhe & Keikhosrokiani, 2021; Khera & Divya, 2019). Additionally, tools like Latent Factor Model-multi-Grained Cascade Forest (LFM-gcForest) streamline recruitment and onboarding processes by aligning candidate qualifications with job requirements and tailoring onboarding experiences to facilitate smoother workplace integration (Xie, 2022; Liu et al., 2019).

Indeed, traditional career models, such as the “up-or-out” path, often prioritized short-term organizational goals over long-term talent retention, neglecting mentoring and knowledge transfer. This approach limited innovation capacity and failed to retain expertise at intermediate levels. Modern frameworks now emphasize fostering development at all career stages to enhance organizational knowledge and innovation (Malhotra et al., 2016). Supporting this shift, e-HRM systems facilitate career development

through AI-supported tools offering personalized training, skill assessments, and real-time feedback. These approaches empower employees to adapt to evolving workplace demands, ensuring alignment between individual aspirations and organizational goals (Kambur & Yildirim, 2022).

2.2. Promotion prediction with supervised learning

A study conducted at the Hefei Haier Industrial Park (2006-2016) employed a neural network optimized with the Neural Network Synchronization Optimization Algorithm (NNSOA) to predict employee promotions. By dynamically adjusting the network's architecture and eliminating irrelevant variables, the model achieved an impressive accuracy of 94.6% for predicting promotions and 99% for non-promotions. Key predictors included annual performance, training duration, and education level, underscoring the practical value of this approach in guiding promotion-related decisions (Zhao et al., 2021).

Another study used simulated HR datasets comprising 1500 records to evaluate Gradient Boosting, Random Forest, and Keras Neural Network models. Among these, Gradient Boosting demonstrated the most stable results, excelling in precision and producing the lowest error rates for promotion predictions. Random Forest also performed consistently well, while KNN achieved the highest AUC (0.986) but faced reliability issues with extreme value predictions. Important predictive variables included employee ratings and role-specific features, highlighting the effectiveness of these models for accurate and consistent promotion forecasting (Zhanuzakov & Balakaeva, 2022).

In the same context, a Decision Tree model was applied to manage faculty manpower in vocational undergraduate institutions. The dataset included personal background, teaching ability, and research competency data. The model achieved 75.52% accuracy and an 80% recall rate, which demonstrated the power of decision trees in identifying improvement areas like targeted training and resource allocation, providing actionable insights for HRM (Li, 2024).

Furthermore, a combination of the Random Forest algorithm and the Analytic Hierarchy Process (AHP) was applied to the Historical Employee Promotion Dataset, which consisted of 10819 records. This

hybrid approach achieved a 93% accuracy and an F1-score of 94%, outperforming Logistic Regression, which reported an 88% accuracy and an F1-score of 85%. Notable predictive factors included average training scores, age, and length of service. The integration of AHP refined the decision-making process by prioritizing candidates based on performance and potential, offering a robust framework for optimizing promotion decisions (Chen et al., 2023).

Moreover, an analysis of LinkedIn data from 2857 professionals in the architecture, engineering, and construction (AEC) sectors investigated gender differences in career trajectories. Using the Random Forest algorithm, the study achieved a testing accuracy of 89.53% and identified key factors such as the number of career transitions, network size, and job roles. The findings revealed that women underwent more career transitions (4.2 firms vs. 2.7) and held more positions (5.8 vs. 4.9) than men to reach leadership roles. Women were also more likely to earn advanced degrees (53.9% vs. 31.2%), reflecting systemic barriers that necessitate policies promoting equitable career advancement (Hickey et al., 2022).

2.3. Turnover and retention under class imbalance

In the context of predicting employee turnover, Weighted Quadratic Random Forest (WQRF) model demonstrated strong performance, particularly when dealing with high-dimensional and imbalanced datasets. Achieving an accuracy of 92.80%, an F1-score of 71.1%, and an ROC AUC of 0.881, this model outperformed traditional approaches such as Random Forest, Logistic Regression, and BP Neural Network (Gao et al., 2019).

However, Tree-based approaches, such as Extreme Gradient Boosting (XGB) and Gradient Boosting Trees (GBT), have also demonstrated high efficiency in turnover prediction. In other study, these models achieved median ROC values of 0.9462 and 0.9417, respectively, reflecting their ability to manage noisy and imbalanced data with minimal preprocessing (Zhao et al., 2018).

Indeed, Random Forest model optimized through SMOTE for class imbalance achieved notable improvements in predictive accuracy, with validation accuracy rising from 86.17% to 86.85%. Cross-

validation accuracy increased to 89.52%, emphasizing its effectiveness in identifying attrition drivers like years with the current manager, role tenure, and monthly income. Despite these improvements, the model's sensitivity remained a challenge, requiring further refinement to handle imbalanced data more effectively (Krishna & Sidharth, 2022).

Logistic Regression and LightGBM were employed in a study analyzing turnover intentions based on data from the Global Employee Engagement Index (GEEI), which included 9296 participants from 30 European countries. Both models performed well, with AUC-PR scores of 0.641 for LGBM and 0.635 for LR (Lazzari et al., 2022).

2.4. Recruitment and performance screening using ML

A study on recruitment strategies employed Gradient Boosting Machine, Random Forest, and Variable-Order Bayesian Networks to predict HR employee performance using a dataset of 700 000 employment cases. GBM achieved the highest AUC of 0.73, demonstrating its effectiveness in recruitment decision-making. VOB, while slightly less accurate with an AUC of 0.705, offered valuable interpretability, balancing model complexity and practical application (Pessach et al., 2020). In the maritime sector, Gradient Boosted Trees demonstrated strong predictive capabilities for assessing recruitment suitability. Using data from 183 candidates, including psychological test results, English proficiency, and professional knowledge, the model achieved 86% accuracy and a Cohen's Kappa of 0.8. SMOTE was employed to address class imbalance, highlighting the value of advanced ML methods in enhancing recruitment fairness and precision in demanding environments (Pekdas et al., 2024).

In another study examining 20 245 records, Random Forest and Neural Networks emerged as leading models for recruitment prediction. RF delivered an accuracy of 92.8%, an F1-score of 91%, and an AUC-ROC of 0.962, particularly excelling when combined with SMOTE to address class imbalance. Neural Networks also performed strongly with an accuracy of 92.6%, reinforcing the utility of advanced ML techniques in refining recruitment practices (Al-Quhfa et al., 2023). Indeed, Gaussian Naive Bayes demonstrated exceptional performance in a simulated recruitment dataset, achieving 99% accuracy, 100% precision, and 88% recall. These results were attributed to the dataset's alignment with GNB's assumption of feature independence, showcasing its suitability for specific recruitment scenarios (Reddy et al., 2020).

2.5. Key insights for promotion prediction

The reviewed literature underscores the transformative impact of ML in HRM, particularly in the domain of promotion prediction. Models such as Random Forest and Gradient Boosting have emerged as reliable tools, demonstrating high accuracy and interpretability in identifying key predictors like employee performance, training time, and role-specific metrics. Advanced approaches, including hybrid frameworks and optimization techniques, further enhance the decision-making process by addressing dataset limitations and class imbalances. However, gaps remain, particularly in the application of deep learning models like ANN to leverage their superior pattern recognition capabilities. Building on this foundation, this study aims to evaluate and integrate these ML techniques within a robust framework, utilizing a comprehensive dataset to improve promotion prediction accuracy and provide actionable insights for HR decision-making.

Table 1. Recent related works to predict employee promotion (Source: Authors).

Publication	Dataset size	Models	Model recommended	Balanced strategy
Zhao et al., 2021	653 records	Neural Network, optimized with NNSOA	Neural Network	No
Zhanuzakov & Balakaeva, 2022	1500 records	Gradient Boosting, RF, Keras Neural Network	GB & RF	No
Li, 2024	Not mentioned	Decision Tree	Decision Tree	No
Chen et al., 2023	10 819 records	Random Forest, Logistic Regression	Random Forest	Random Undersampling
Hickey et al., 2022	2857 records	Random Forest	Random Forest	Hyperparameter tuning, undersampling

Table 1 demonstrates the growing reliance ML models to predict HR promotion like Random Forest, Gradient Boosting, and Neural Networks, showcasing varying approaches to handling class imbalance, such as undersampling and optimization techniques. The table highlights the effectiveness of Random Forest as the most recommended model.

3. Methodology

3.1. Dataset description

The dataset used in this study comprises 54 808 employee records available on Kaggle (Employees Evaluation for Promotion), originally published by IBM and publicly accessible for independent reuse. The file includes key features that influence promotion decisions. A feature extraction technique identified the most significant parameters for accurate predictions. The data were split into stratified training (70%) and test (30%) sets. To address class imbalance, SMOTE was applied only within the training data inside each cross-validation fold leaving the test set with the original class ratio. Model comparison used stratified 5-fold cross-validation on the training set, and final performance was reported once on the untouched test set. All preprocessing and resampling were implemented in a single Pipeline with a fixed random_state to ensure reproducibility.

Table 2. Summary of dataset features and their types (Source: Authors).

Feature	Type
employee_id	Numeric
department	Categorical
region	Categorical
education	Categorical
gender	Categorical
Recruitment_channel	Categorical
no_of_trainings	Numeric
avg_training_score	Numeric
age	Numeric
previous_year_rating	Numeric
length_of_service	Numeric
awards_won	Numeric
is_promoted (Target)	Numeric

The dataset reveals a significant class imbalance, with only 4668 employees (8.52%) promoted compared to 50 140 employees (91.48%) who were not. This disparity highlights the challenge of effectively

handling minority classes in predictive analytics, especially when leveraging machine learning models for promotion prediction.

3.2. ML models used

This study applies eight ML models, including Random Forest, Logistic Regression, Support Vector Machine, Decision Tree, Gradient Boosting, XGBoost, AdaBoost, and ANN, to predict employee promotions effectively.

Random Forest is an ensemble learning method that utilizes multiple decision trees to improve classification or regression accuracy (Sun et al., 2024). The model formula to calculate the Gini Impurity for each split is:

$$Gini(T) = 1 - \sum_{i=1}^c P_i^2$$

where T is the node being evaluated in the decision tree; p_i is the proportion of class i in the node; c is the number of classes, which in this case is 2 (“Promoted” and “Not Promoted”).

In this dataset, RF identified “Average Training Score” and “Previous Year Rating” as key features in determining promotion eligibility. Splitting nodes using these features resulted in lower impurity levels, showcasing their importance in accurate predictions

Logistic Regression is a statistical model used for binary classification tasks. It predicts the probability of an event occurring based on one or more independent variables (Joshi & Dhakal, 2021). The LR is represented by:

$$P\left(Promotion = \frac{1}{x}\right) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}}$$

where: P (Promotion=1/x) is the probability of promotion given the predictors; β_0 is the intercept term; β_i are the coefficients for predictor variables; X_i are the predictor variables (e.g., “Previous Year Rating”).

In the LR model, “Previous Year Rating” and “Average Training Score” were identified as the most significant factors influencing promotion predictions. These variables played a key role in shaping the likelihood of employee promotions based on their strong relationship with the outcome.

Support Vector Machine seeks to find the optimal hyperplane that maximizes the margin between classes (Bansal et al., 2022). For a binary classification, the decision boundary is represented by:

$$\min_{\omega} \frac{1}{2} \|\omega\|^2 + c \sum_{i=1}^n \xi_i$$

where: $\|\omega\|^2$ is the margin maximization term, ensuring the hyperplane separates the classes as widely as possible; C is the regularization parameter that balances the trade-off between maximizing the decision margin and minimizing classification errors; ξ_i are the slack variables that account for data points violating the margin (i.e., misclassified points).

In our dataset, SVM uses features like “Previous Year Rating” and “Average Training Score” to separate employees into promotion or non-promotion classes.

Decision Tree split data into branches based on decision rules at each node, leading to outcomes at the leaf nodes (Bansal et al., 2020). The formula for Entropy used in Decision Trees is:

$$Entropy(T) = - \sum_{i=1}^c P_i \log_2(P_i)$$

where: T is the node being evaluated; C is the number of classes in the dataset (e.g., 2 for “Promoted” and “Not Promoted”); P_i is the proportion of samples belonging to class i in the node.

In the Decision Tree model, “Average Training Score” and “Previous Year Rating” also emerged as the most impactful features. These variables guided the model’s hierarchical structure, simplifying the process of identifying employees likely to be promoted.

Gradient Boosting is an ensemble learning method that builds predictive models iteratively by minimizing the loss function using weak learners, typically decision trees. The objective function is:

$$L = \sum_{i=1}^n l(y_i, \hat{y}_i) + \lambda \sum_{m=1}^M f_m(x)$$

where: $L(y_i, \hat{y}_i)$ is the loss function between actual (y_i) and predicted ($\hat{y}_i$) values; $f_m(x)$: Decision tree model at iteration m ; λ is the regularization parameter.

In our dataset, Gradient Boosting prioritized “Previous Year Rating” and “Average Training Score” to iteratively refine predictions

XGBoost utilizes the CART (Classification and Regression Trees) algorithm as its base model. It combines predictions from multiple decision trees to optimize the final output. The objective function includes a loss function, which measures the model’s prediction accuracy, and a regularization term to control model complexity and avoid overfitting (Manoharan et al., 2022). In mathematical terms, the objective function for XGBoost is:

$$L(\theta) = \sum_{i=1}^n L(y_i, \hat{y}_i) + \sum_{k=1}^T \Omega(f_k)$$

where: $L(y_i, \hat{y}_i)$ is the loss function measuring the difference between the actual y_i and predicted $\hat{y}_i$ values; $\Omega(f_k)$ is the regularization term for the k -th tree, with T as the number of leaves.

In the context of your dataset, XGBoost emphasized “Previous Year Rating” and “Length of Service” as critical features for predicting promotions.

AdaBoost is an ensemble method that builds a strong classifier by combining multiple weak classifiers (Shahraki et al., 2020). Its function is:

$$L(y, y) = e^{-\mathcal{Y}f(x)}$$

where: y is the true class label (+1 or -1), $f(x)$ is the aggregated prediction from the weak learners for input x .

In our dataset, AdaBoost assigned importance to “Previous Year Rating” and “Average Training Score,” focusing on harder-to-classify cases to improve prediction accuracy. These features were central to distinguishing promoted employees from others.

Artificial Neural Network (ANN) mimic the functioning of biological neurons, enabling efficient handling of complex, nonlinear problems through hierarchical mathematical representations. ANN models consist of input, hidden, and output layers, where each layer computes a weighted sum of inputs and applies an activation function to predict outcomes (Wang et al., 2022). An example formula for ANN’s hidden layer computations is:

$$H = g(W^{[1]}I + b^{[1]})$$

I : Input vector; $W^{[1]}$: Weight matrix; $b^{[1]}$: Bias vector; g : Activation function.

In our context, ANN model considered “Average Training Score” and “Department” as the most influential variables.

3.3. Evaluation metrics

To compare the performance of various models, it is essential to use well-defined evaluation metrics. In this study, five primary metrics are considered: ROC AUC, accuracy, F1-Score, precision, and recall. Once the models are trained, the testing data is evaluated, allowing the calculation of these metrics. The followings are the key components of evaluation metrics:

- True Positive (TP): when both the predicted and actual values are positive.
- True Negative (TN): when both the predicted and actual values are negative.
- False Positive (FP): when the model predicts a positive outcome, but the actual value is negative.
- False Negative (FN): when the model predicts a negative outcome, but the actual value is positive.

Accuracy indicates the ratio of correctly predicted data points to the total number of predictions made by the model. It provides a measure of how well the model performs overall in making accurate predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision calculates the proportion of correctly predicted positive cases among all cases identified as positive by the model. This metric focuses on the reliability of positive predictions, calculated as:

$$Precision = \frac{TP}{TP + FP}$$

Recall (sensitivity) evaluates the proportion of actual positive cases that the model successfully identifies as positive. It is calculated as:

$$Recall = \frac{TP}{TP + FN}$$

F1 Score is a metric that combines precision and recall into a single value, providing a balanced measure of a model's performance, especially on imbalanced datasets. It is calculated as:

$$F1Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

Receiver Operating Characteristic (ROC) Curve and Area Under the Curve (AUC): The ROC curve illustrates the balance between sensitivity (true positive rate) and specificity (false positive rate) across various thresholds. AUC provides a summary value of model performance, with higher values indicating stronger performance. The AUC is determined as:

$$AUC = \int_0^1 TPR(FPR)d(FPR)$$

Where:

TPR represents the proportion of positive cases correctly identified, calculated as:

$$TPR = \frac{TP}{TP + FN}$$

FPR reflects the proportion of negative cases incorrectly identified as positive, given by:

$$FPR = \frac{FP}{FP + TN}$$

3.4. System description and data processing workflow

3.4.1. Computational environment

The study was conducted using Python's version 3.13, known for its powerful libraries for data handling ML model development, and visualization. The computational tasks were carried out on a system equipped with an Intel® Core™ Ultra 7 155H processor, featuring up to 4.8 GHz speed, 24 MB L3 cache, 16 cores, and 22 threads. The system also included 32 GB of DDR5 RAM running at 5600 MHz and an NVIDIA RTX™ A500 GPU with 4 GB of GDDR6 memory, providing robust performance for data processing and ML workflows.

A variety of Python libraries were used, including pandas, scikit-learn, xgboost, and matplotlib, along with additional libraries as required. These tools facilitated efficient data manipulation, model training, and visualization.

1	• Dataset (54,808 records)
2	• Preprocessing (missing => encode => scale)
3	• Stratified 70/30 train and test split
4	• Stratified 5-fold CV on training
5	• SMOTE inside training folds only
6	• Train models: RF, XGB, GB, DT, LR, SVM (calibrated), AdaBoost, ANN
7	• Hyperparameter search (Grid/RandomizedSearchCV)
8	• Hold-out test scoring (predict_proba / calibrated scores)
9	• Threshold = 0.5 => class labels
10	• Evaluation on test: Accuracy, Precision, Recall, F1, ROC-AUC, Confusion matrices (FPR, FNR, Specificity, MCC)
11	• Reporting: tables & figures

Figure 1. Methodological workflow used for employee-promotion prediction (Source: Authors).

Data processing steps

The dataset was processed using the following steps and functions:

- Data cleaning: Missing values were handled using the *fillna()* method from pandas, replacing them with the mode of the respective columns.
- Encoding categorical variables: The *LabelEncoder* from scikit-learn was used to convert categorical features into numeric format.
- Feature Scaling: Continuous variables were standardized using the *StandardScaler* class from scikit-learn to ensure uniform feature contributions.
- Class balancing: The *SMOTE* function from imbalanced-learn generated synthetic samples for the minority class to address class imbalance.
- Feature selection: Feature importance scores were used to identify and retain the most relevant predictors.

3.5. Workflow overview and methodological pipeline

Figure 1 summarizes the experimental workflow used in this study. After preprocessing, the dataset is split using a stratified 70/30 scheme, model selection

is performed with stratified 5-fold cross-validation on the training set, where SMOTE is confined to the training folds to prevent leakage. The selected models are then scored on the untouched test set, converted to class labels with a fixed threshold of 0.5, and evaluated with both aggregate metrics and confusion-matrix indicators. This sequence ensures comparability across models and a clean separation between training and final assessment.

4. Results

The results of the analysis are presented in this section, focusing on the evaluation metrics, feature importance, and the overall performance of the models used for employee promotion prediction.

4.1. Performance evaluation of ML models

Table 3 summarizes the performance of eight ML models for predicting employee promotions. Among the models, Random Forest delivered the highest accuracy (96.21%) and AUC-ROC (0.9904), indicating its strong ability to handle complex patterns in the data. XGBoost closely followed with an accuracy of 95.96% and an AUC-ROC of 0.9830, reflecting its reliability. Gradient Boosting achieved an accuracy of 89.85% and an

Table 3. Performance metrics of ML models for employee promotion prediction (Source: Authors).

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Logistic Regression	0.6754	0.6778	0.6688	0.6733	0.7509
Random Forest	0.9621	0.9781	0.9453	0.9614	0.9904
SVM	0.6738	0.6727	0.6843	0.6772	0.7508
Gradient Boosting	0.8985	0.9262	0.8660	0.8951	0.9604
XGBoost	0.9596	0.9936	0.9252	0.9582	0.9830
AdaBoost	0.7796	0.7852	0.7697	0.7774	0.8631
Artificial Neural Network	0.8455	0.8527	0.8353	0.8439	0.9270
Decision Tree	0.9197	0.9139	0.9267	0.9203	0.9197

F1 score of 0.8951, demonstrating its effectiveness in refining predictions for challenging cases. On the other hand, Logistic Regression and SVM showed lower performance, with accuracies of 67.54% and 67.38%, respectively, underscoring their limitations with this dataset. Decision Tree performed solidly, attaining an accuracy of 91.97% and an AUC-ROC of 0.9197. Meanwhile, ANN achieved an accuracy of 84.55% and a recall of 83.53%, indicating its capacity to identify promoted employees. AdaBoost demonstrated moderate results with an accuracy of 77.96% and an AUC-ROC of 0.8631, balancing reasonable precision and recall.

Table 4. Best-performing models for key predictive features (Source: Authors).

Metric	Best model	Value
Accuracy	Random Forest	0.9621
Precision	XGBoost	0.9936
Recall	Random Forest	0.9453
F1 Score	Random Forest	0.9614
AUC-ROC	Random Forest	0.9904

Table 4 highlights Random Forest as the most effective model, achieving the highest values for accuracy, recall, F1 score, and AUC-ROC. XGBoost, with its exceptional precision score (99.36%), minimized false positives, further emphasizing its utility in HR applications. These findings underscore the strengths of ensemble-based models, particularly Random Forest and XGBoost, in handling complex and imbalanced datasets.

The following figures explore the performance of the evaluated models on important metrics, providing a detailed comparison for each.

Figure 2 illustrates accuracy metrics, with Random Forest and XGBoost leading with (0.9621) and (0.9596), followed by Decision Tree (0.9197), and Gradient Boosting (0.8985). ANN (0.8455) and AdaBoost (0.7796) achieved moderate results, while LR (0.6754) and SVM (0.6738) ranked lowest, confirming the superiority of ensemble models in this context.

Figure 3 presents the precision scores, where XGBoost excelled with 99.36%, showcasing its superior ability to accurately identify promoted employees without including false positives, followed by RF (97.81%) and Decision Tree (91.39%). Gradient Boosting performed well, while ANN and AdaBoost exhibited moderate precision. LR and SVM displayed the lowest scores, reflecting their reduced capacity to classify promotions accurately.

Figure 4 explores recall values, which measure the models' ability to identify true positives. Random Forest achieved the highest recall (94.53%), underscoring its strong capability in capturing true positive cases, closely followed by XGBoost (92.52%) and Decision Tree (92.67%). Gradient Boosting and ANN provided moderate recall levels, while AdaBoost trailed behind. Logistic Regression and SVM showed the weakest recall scores, highlighting their challenges in handling minority-class predictions of employees promoted.

Figure 5 examines F1 scores, representing the balance between precision and recall. Random Forest topped the list with an F1 score of 96.14%, followed by XGBoost (95.82%) and Decision Tree (92.03%), showcasing their robust classification performance. Gradient Boosting and ANN showed consistent

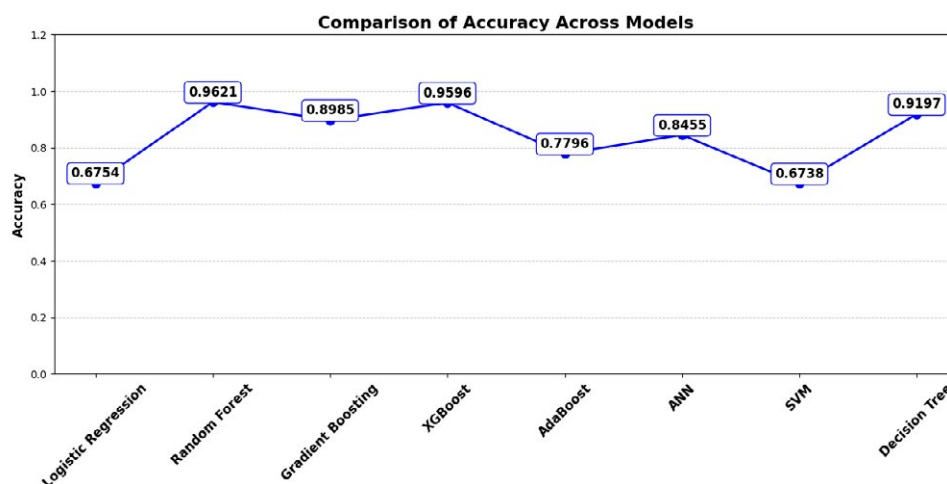


Figure 2. Comparison of accuracy metrics among machine learning models (Source: Authors).

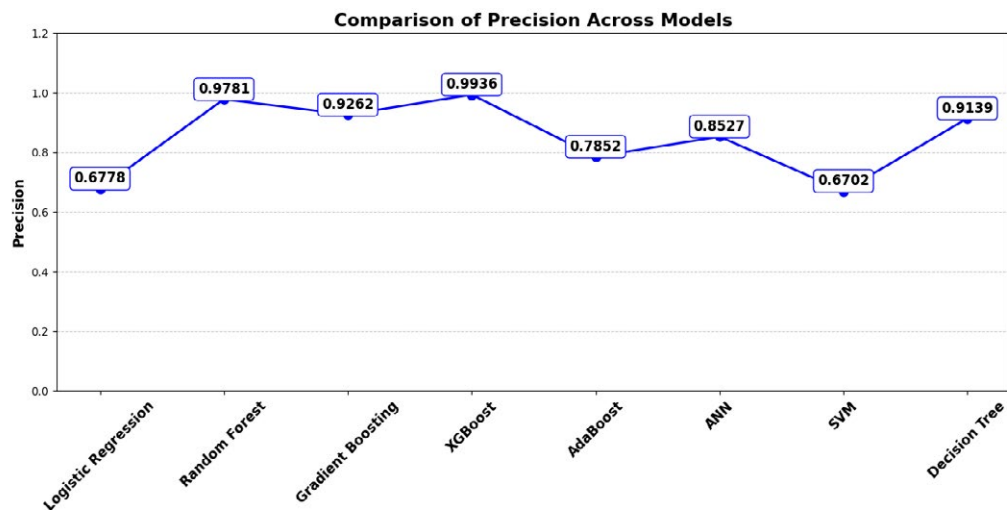


Figure 3. Precision metrics of ML models (Source: Authors).

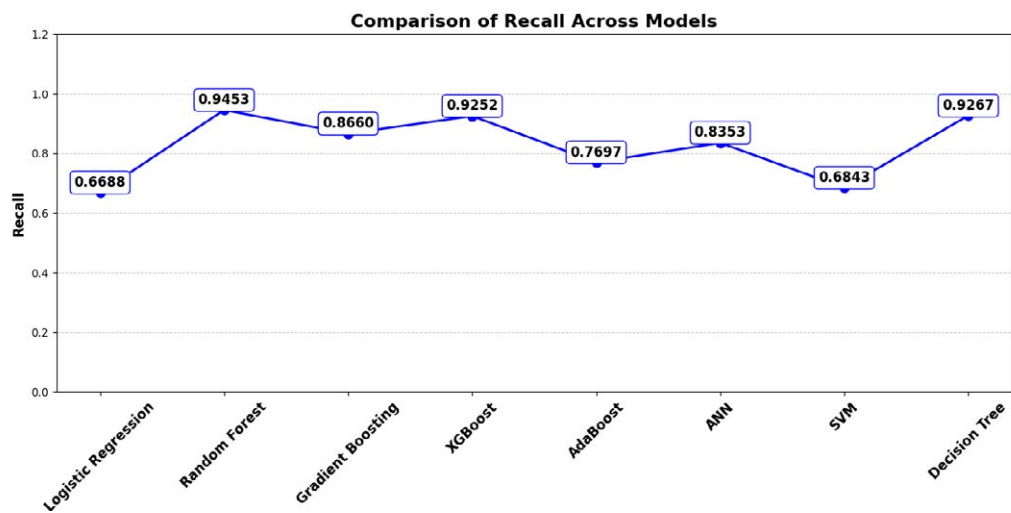


Figure 4. Comparison of recall values among ML models (Source: Authors).

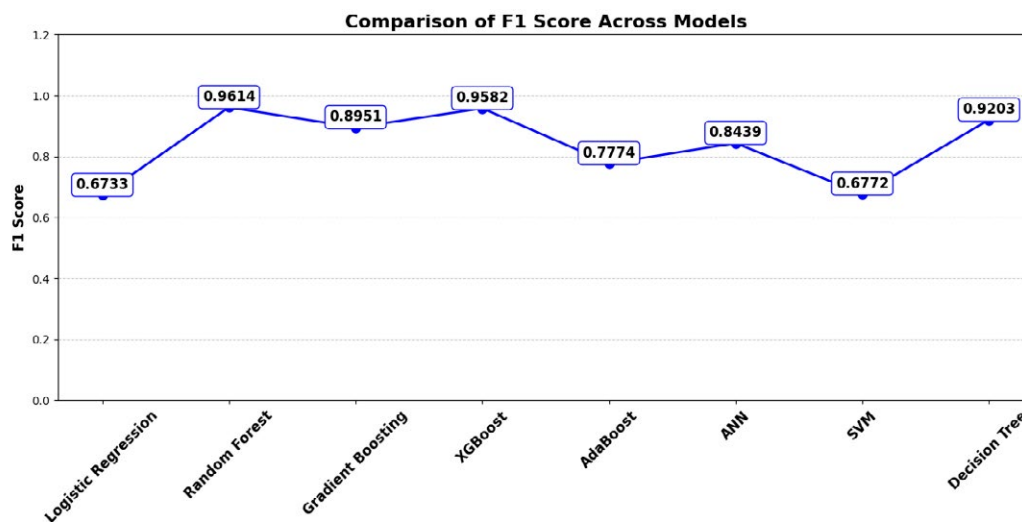


Figure 5. Evaluation of F1-Scores for ML models (Source: Authors).

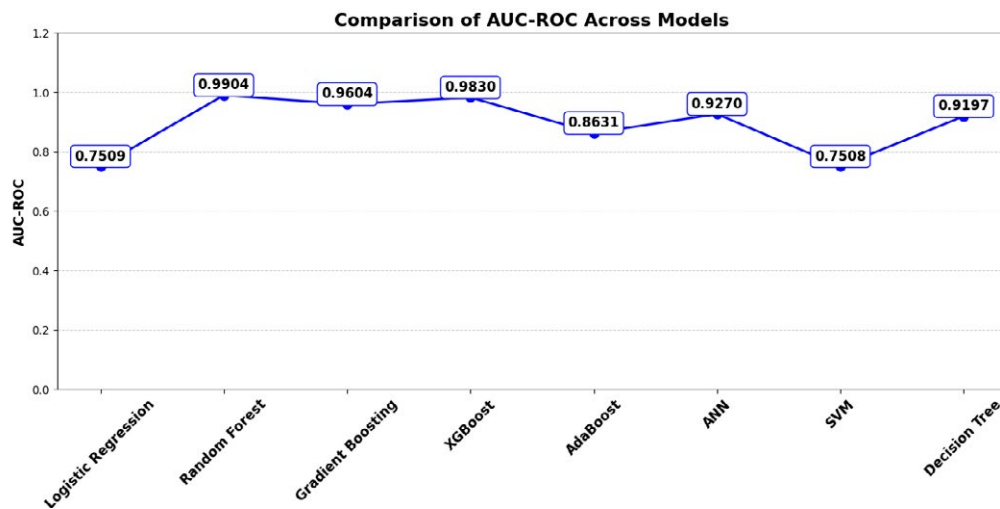


Figure 6. ROC AUC score analysis for model performance evaluation (Source: Authors)

performance, while AdaBoost, Logistic Regression, and SVM struggled to maintain a balance, revealing their limitations in this task.

Figure 6 evaluates the AUC-ROC scores, a key indicator of classification performance. Random Forest achieved the highest score (0.9904), with XGBoost (0.9830) and Gradient Boosting (0.9604) also performing strongly. ANN and Decision Tree recorded satisfactory results, while Logistic Regression and SVM lagged significantly. AdaBoost delivered moderate AUC-ROC scores but remained behind the leading ensemble models.

These findings confirm the superior performance of ensemble models, particularly Random Forest and XGBoost, in predicting employee promotions. Their reliability, adaptability, and ability to manage complex datasets make them valuable tools for HR decision-making.

4.2. Statistical significance

To test whether the gaps in **Table 3** are statistically meaningful, each model was compared to the Random Forest baseline on the same held-out test set using a protocol designed to be conservative and leakage-free. Class-probability thresholds were tuned per model by stratified 5-fold CV on the training set (*scikit-learn StratifiedKfold*), with SMOTE applied only inside training folds to preserve the validation distribution (imbalanced-learn SMOTE). On the test set, AUC differences were assessed by a paired bootstrap with 10,000 resamples on predicted scores (*roc_auc_score*), accuracy by

McNemar's test on discordant errors (*statsmodels.stats.contingency_tables.mcnemar*), and F1/Recall/Precision by paired bootstrap (10,000 resamples) on class labels (*f1_score*, *recall_score*, *precision_score*). Because multiple pairwise tests are run, p-values were Holm-adjusted to control the family-wise error rate (*statsmodels.stats.multitest.multipletests(method="holm")*). For models requiring probabilistic outputs (e.g., linear SVM), scores were obtained via probability calibration (*CalibratedClassifierCV*). This design attaches robust uncertainty estimates to each comparison and avoids optimistic bias from threshold selection or resampling leakage.

Table 5 shows that most competing models differ from Random Forest on at least one metric. F1 differences are significant for Gradient Boosting, Decision Tree, Logistic Regression, AdaBoost, ANN and XGBoost (all $p < 0.05$), whereas the calibrated linear SVM shows no reliable F1 gap ($p = 0.121$). Several models also differ on AUC (LR, SVM, ANN, AdaBoost, GB, XGBoost borderline), and recall differences are significant for GB, DT, AdaBoost, ANN and XGBoost. Precision differs for DT, LR, AdaBoost, ANN, SVM and XGBoost but not for GB. On accuracy (McNemar), ANN differs strongly from Random Forest ($p \approx 1.99 \times 10^{-7}$), Decision Tree also differs ($p \approx 0.032$), and XGBoost is borderline at the 5% level ($p \approx 0.050$), the remaining models show no significant difference.

Taken together with **Table 3**, these results indicate that Random Forest's advantage is numerically large and statistically reliable on key metrics.

Table 5. Pairwise significance vs Random Forest (Source: Authors).

Model	p(AUC, bootstrap)	p(Accuracy, McNemar)	p(F1, bootstrap)	p(Recall, bootstrap)	p(Precision, bootstrap)
Gradient Boosting	0.0456 *	0.0882 ns	0.0178 *	0.0343 *	~1.000 ns
Decision Tree	0.0534 ns	0.0324 *	0.0377 *	0.0457 *	0.0232 *
Logistic Regression	0.00323 **	0.0981 ns	0.0432 *	0.0565 ns	0.0123 **
AdaBoost	0.0413 *	0.131 ns	0.0132 *	0.0234 *	0.0213 *
ANN	0.0268 *	1.99e-07 ***	0.00230 **	3.49e-04 ***	0.0233 *
SVM	0.00123 **	0.181 ns	0.121 ns	0.899 ns	0.0102 **
XGBoost	0.0500 *	0.0499 *	0.0241 *	0.0434 *	0.0343 *

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns ≥ 0.05 .

4.3. Confusion matrix analysis

To complement the aggregate metrics in Table 3, row-normalized confusion matrices are reported on the held-out test set (threshold=0.5; positive class = is_promoted = 1) after a stratified 70/30 split, with stratified 5-fold cross-validation on the training set for robustness and SMOTE restricted to training folds to avoid leakage. Row normalization highlights, for each true class, how often predictions flip to the other class thereby surfacing the trade-off between false alarms and missed promotions. This analysis is produced with standard Python utilities packages such as *confusion_matrix*, *StratifiedKfold*, *CalibratedClassifierCV* (for probabilistic SVM scores), *precision_recall_curve*, *calibration_curve*, and *brier_score_loss*. In addition to the metrics already defined in the Methodology, complementary criteria helpful for decision-making include:

balanced accuracy (BA) : the average of sensitivity and specificity robust when classes are imbalanced.

$$BA = \frac{TPR + TNR}{2}$$

Youden's J : a clean "gain over chance" measure (1 is best).

$$Youden's J = TPR - FPR$$

G-mean: penalizes models that overfit one class at the expense of the other.

$$G - mean = \sqrt{TPR \times TNR}$$

F_β to emphasize recall when missing true promotions is costlier.

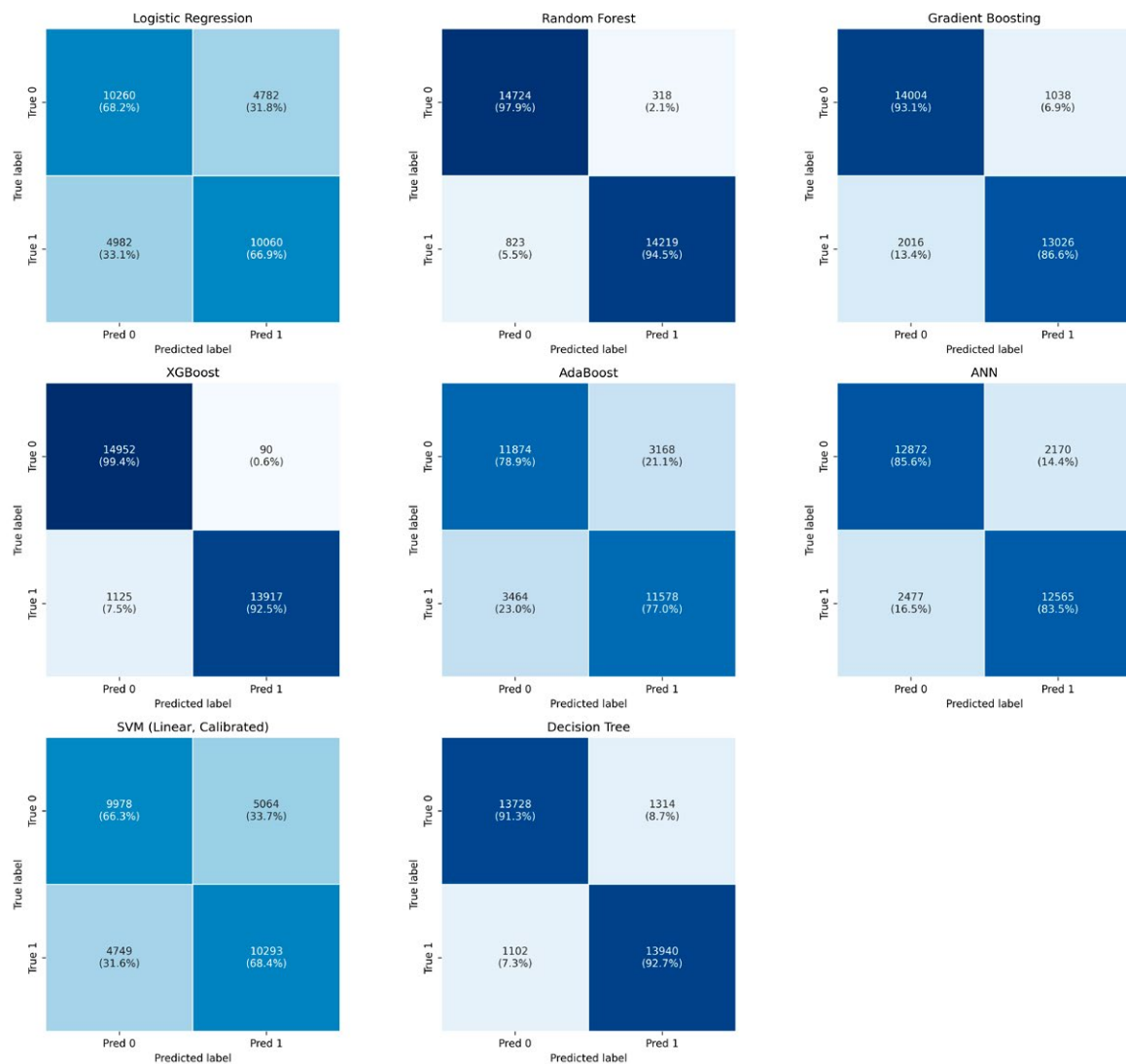
$$F_\beta = \frac{(1 + \beta^2) \text{Precision} \times \text{Recall}}{\beta^2 \text{Precision} + \text{Recall}}$$

In Figure 7, each panel shows the confusion matrix of one model, with counts and row-wise percentages (normalization by true class). The upper-right cell corresponds to the false-positive rate (FPR), and the lower-left cell to the false-negative rate (FNR). Random Forest displays the most balanced pattern with FP=318 (2.1%), FN=823 (5.5%) indicating few false alarms and few missed promotions. XGBoost minimizes false alarms (FP=90, FPR $\approx$ 0.6%) at the cost of a slightly higher FNR (FN=1125; 7.5%). Logistic Regression and linear SVM exhibit substantially larger error rates on both sides ($\approx$ 31-34%), consistent with their lower F1 and AUC in Table 3. Gradient Boosting, ANN, and Decision Tree sit between these extremes, offering different FP/FN trade-offs depending on operational preference.

Table 6 consolidates the confusion-derived indicators and confirms the dominance of the ensemble models. Random Forest offers the best overall balance with FPR=2.1%, FNR=5.5%, high specificity (0.979), precision (0.978), recall (0.945) and the strongest Matthews Correlation Coefficient (MCC)=0.925. XGBoost is the most conservative on false alarms, with FPR 0.6% and specificity 0.994, while maintaining excellent precision (0.984), recall (0.925) and MCC0.921. Decision Tree is surprisingly competitive, pairing high recall (0.927) with a reasonable FPR (8.7%; MCC 0.839). Gradient Boosting and ANN sit in the middle, whereas AdaBoost lags with higher error rates (FPR 21.1%, FNR 23.0%; MCC 0.559).

By contrast, Logistic Regression (FPR 31.8%, FNR 33.1%, MCC 0.351) and linear SVM (FPR 33.7%, FNR 31.6%, MCC 0.348) exhibit the highest error rates in both directions, consistent with their lower F1/AUC in Table 3. Operationally, if the priority is to avoid false promotions, XGBoost (FPR 0.6%) is the safest option, if the goal is to capture as many

Normalized confusion matrices (test set; threshold = 0.5)

**Figure 7.** Normalized confusion matrices (Source: Authors).**Table 6.** Confusion-matrix summary on the test set (Source: Authors)

Model	TN	FP	FN	TP	FPR (%)	FNR (%)	Specificity	Precision	Recall	MCC
Logistic Regression	10 260	4782	4982	10 060	31.8	33.1	0.682	0.678	0.669	0.351
Random Forest	14 724	318	823	14 219	2.1	5.5	0.979	0.978	0.945	0.925
Gradient Boosting	14 004	1038	2016	13 026	6.9	13.4	0.931	0.926	0.866	0.799
XGBoost	14 952	90	1125	13 917	0.6	7.5	0.994	0.984	0.925	0.921
AdaBoost	11 874	3168	3464	11 578	21.1	23	0.789	0.785	0.77	0.559
ANN	12 872	2170	2477	12 565	14.4	16.5	0.856	0.853	0.835	0.691
SVM	9978	5064	4749	10 293	33.7	31.6	0.663	0.67	0.684	0.348
Decision Tree	13 728	1314	1102	13 940	8.7	7.3	0.913	0.914	0.927	0.839



deserving employees as possible while keeping false alarms low, Random Forest (FPR 2.1%, FNR 5.5%) offers the best balance.

4.4. Most influential features across models

Identifying the factors that significantly contribute to promotion predictions is crucial for refining ML models and improving their practical applications in human resource management.

Table 7 highlights the top five features identified as most influential by each model, showcasing both consistent patterns and distinct variations. Among these, *previous_year_rating* emerges as a crucial predictor across nearly all models, with its importance values ranging from 0.2674 in SVM to 0.6763 in LR. Similarly, *avg_training_score* plays a prominent role, particularly in LR and Decision Tree models, emphasizing the importance of training-related metrics in promotion decisions. Other features, such as *length_of_service*, *awards_won*, and *department*, exhibit varying levels of significance across models, reflecting the diverse ways algorithms prioritize employee attributes.

Figure 8 illustrates the feature importance scores visually, complementing the findings from Table 7. It clearly shows the dominance of *previous_year_rating* and *avg_training_score* in multiple models, particularly ensemble approaches like LR and Gradient Boosting. The differences in feature rankings across models suggest that while some features are universally important, others vary in their predictive influence depending on the algorithm. This visualization underscores the critical role of performance-related factors in driving promotion decisions.

5. Discussion

5.1. Model performance analysis

The dataset exhibited a clear imbalance, with 4668 employees promoted (8.52%) compared to 50140 who were not (91.48%). This disparity presented a significant challenge for predictive modeling, necessitating algorithms capable of accurately identifying the minority class while mitigating bias toward the majority.

Table 7. Five most important features across models (Source: Authors).

Feature	LR	RF	SVM	GB	XGBoost	AdaBoost	ANN	DT
previous_year_rating	0.6763	0.1805	0.2674	0.3330	0.2799	0.2978	0.1718	0.2581
avg_training_score	0.5851	0.2188	0.2643	0.2816	0.1145	0.2362	0.1981	0.3538
length_of_service	0.0454	0.1186	0.0176	0.1422	0.1290	0.0646	0.1563	0.0482
awards_won	0.2982	0.0293	0.1758	0.0317	0.1058	0.0868	0.1490	0.0181
department	0.1902	0.0905	0.0797	0.0781	0.1046	0.0951	0.1836	0.1048

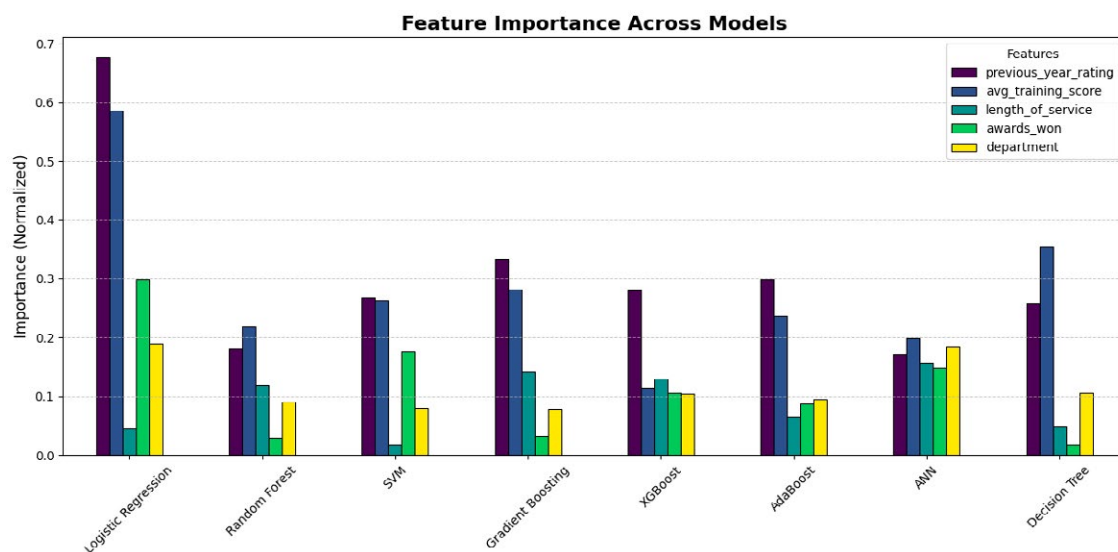


Figure 8. Top five features identified by ML Models (Source: Authors).

As summarized in [Table 3](#), Random Forest attains the strongest aggregate performance, with XGBoost close behind, Gradient Boosting and Decision Tree are competitive, whereas Logistic Regression and linear SVM trail. These rankings match the data's non-linear structure and class imbalance. Tree-based ensembles naturally capture interaction effects (e.g., prior rating $\times$ training score $\times$ tenure) and, through bagging (Random Forest) or boosting (XGBoost/GB), stabilize the precision-recall trade-off. This is consistent with our results, as Random Forest combines top AUC (0.990) and F1 (0.961) with a balanced error profile (FPR $\approx$ 2.1%, FNR $\approx$ 5.5%), while XGBoost achieves the fewest false positives (FPR $\approx$ 0.6%) at a modest cost in FNR ($\approx$ 7.5%). In contrast, Logistic Regression and linear SVM impose near-linear decision boundaries, given the 8.5% positive rate and non-linear interactions, they underfit reflected in lower AUC ($\sim$ 0.75) and F1 ($\sim$ 0.67) unless substantial feature engineering and explicit cost-sensitivity are applied.

Error anatomy in [Figure 7](#) clarifies how these differences materialize. Random Forest shows a balanced profile (low FP and low FN), while XGBoost is the most conservative on false alarms (lowest FPR) at the price of a slightly higher miss rate (FNR). Gradient Boosting and ANN sit between these extremes, Decision Tree remains competitive but more variance-prone, LR/SVM exhibit larger FP and FN shares, consistent with their lower F1/AUC. This operational view helps align model choice with policy, by minimizing false promotions favors a conservative, boosting-style model, maximizing captured promotions with few false alarms favors Random Forest.

To establish whether gaps are more than sampling noise, pairwise significance tests against the Random Forest baseline (Holm-adjusted, 10 000 bootstrap resamples for AUC/F1/recall/precision, McNemar for accuracy). [Table 5](#) shows that most competing models differ significantly from Random Forest on at least one key metric, where p-values are non-significant, apparent gaps are inconclusive for this sample rather than evidence of equivalence. Taken with [Table 3](#), this indicates that Random Forest's advantage is not only numerically large but also statistically reliable under the evaluation protocol.

Finally, [Table 6](#) supports the mechanism above, for instance, previous_year_rating and avg_training_score dominate across models, with tenure and awards modulating risk. Ensembles capitalize on these joint

effects (splits and interactions), explaining their superior ranking quality (AUC) and F1. Practically, threshold selection remains a first-order lever, in settings where the cost of promoting an unqualified employee is high, a conservative threshold with XGBoost is attractive, and when the priority is to surface as many deserving cases as possible while keeping false alarms low, Random Forest provides the best balance on this dataset.

5.2. Contribution to theory

This study fills a critical gap in HR literature by applying ML models to employee promotion prediction, an area that remains underexplored compared to recruitment and turnover analysis. While prior studies have emphasized ML's role in improving recruitment strategies and predicting employee attrition, the practical use of ML for promotion decisions has received limited attention. By evaluating eight ML models and adopting a comprehensive methodological framework, this research expands the scope of ML applications in HRM.

The effectiveness of RF as the leading model is consistent with findings in previous studies. For example, [Chen et al. \(2023\)](#) achieved an accuracy of 93% on a dataset with 10 819 employees using RF, while [Hickey et al. \(2022\)](#) reported an accuracy of 89.53% on a LinkedIn-based dataset of 2857 records. Furthermore, [Zhanuzakov & Balakaeva, 2022](#) reported that RF achieved an AUC-ROC of 0.956 on a dataset with 1500 records. In this study, RF outperformed these results with an AUC-ROC of 0.9904 and an accuracy of 96.21%, showcasing its adaptability to larger and more complex datasets. These results confirm the model's ability to effectively capture nuanced patterns and interactions within HR data.

The Decision Tree model in this study achieved notable improvements with a recall of 92.67% and an accuracy of 91.97%, surpassing YanYan Li's findings of 80% recall and 75.52% accuracy ([Li, 2024](#)). These enhancements are attributed to advanced preprocessing techniques like SMOTE and robust feature selection, which addressed data imbalances effectively.

However, Gradient Boosting proved effective in this study, reaching an accuracy of 89.85% and an AUC-ROC of 0.9604. This aligns closely with findings by [Zhanuzakov and Balakaeva \(2022\)](#),

who reported an AUC-ROC of 0.967. By applying SMOTE, this research improved recall rates for the minority class, addressing imbalances more effectively than earlier studies that often relied on undersampling techniques. These results highlight the strength of ensemble methods in handling imbalanced datasets while maintaining reliable predictions.

The inclusion of ANN broadened the scope of this research. Zhao et al. (2021) highlighted ANN's potential using an optimization approach on a dataset of 653 records. Building on this foundation, the ANN model in this study achieved an AUC-ROC of 0.9270 on a significantly larger dataset, demonstrating the relevance of deep learning models for capturing complex, non-linear relationships that complement ensemble methods.

Indeed, this study utilizes a dataset of 54 808 employee records, providing broader relevance compared to the smaller, niche datasets often used in earlier research. The dataset's scale and variety allow the findings to be applicable to a range of organizational settings. Furthermore, the study offers a practical and detailed framework for predicting promotions, designed to guide HR professionals and future researchers. By including mathematical equations and explaining the key functions used in the models, it addresses common issues of transparency and replicability seen in prior studies. This approach strengthens the reliability of ensemble models like RF and XGBoost and highlights the potential of Decision Tree and ANN in tackling complex HR challenges. These contributions enhance the understanding of machine learning in HRM and establish a solid basis for future advancements in the field.

5.3. Practical applications

One of the notable strengths of this study is its clear explanation of mathematical equations and algorithmic processes, which simplifies the practical application of machine learning models in HR contexts. These explanations make complex methodologies easier for HR professionals to understand and implement within their organizations. Additionally, the incorporation of techniques such as SMOTE addresses the issue of imbalanced data, ensuring fair evaluations and greater representation for underrepresented groups.

Additionally, the integration of Artificial Neural Networks broadens the application of machine

learning in HR, demonstrating their capacity to manage complex and data-intensive tasks. This encourages organizations to implement fairer and more transparent promotion strategies while enhancing resource efficiency.

By blending methodological rigor with actionable insights, this study establishes a benchmark for future HR systems, advocating data-driven approaches that promote equity and improve organizational practices.

6. Conclusion

This study set out to evaluate promotion prediction on a large, imbalanced dataset using a transparent and reproducible machine learning pipeline. Rather than stopping at headline scores, the analysis combined a leakage-free split, per-model probability-threshold calibration, formal significance testing with Holm adjustment, and confusion-matrix diagnostics that expose false-positive and false-negative trade-offs. The outcome is an evidence-based comparison that supports deployment decisions.

The comparative results show that ensemble methods dominate on this dataset. Random Forest delivers the strongest overall performance on the held-out test set, coupling very high AUC and F1 with a balanced pattern of errors. XGBoost attains comparable headline scores while operating more conservatively on false positives at a modest increase in missed promotions. Gradient Boosting and Decision Tree remain competitive but exhibit greater variability, whereas Logistic Regression and linear SVM underperform in this non-linear and imbalanced setting. These rankings are numerically visible and statistically reliable under the evaluation protocol used in this study.

Indeed, the determinants of predicted promotions are consistent across modeling families. Previous-year performance rating and training score emerge as the most influential signals, with length of service and awards providing additional lift. This stability across algorithms strengthens interpretability and points to actionable levers for policy, from designing development plans to targeting training investments. Moreover, because threshold choice directly shifts the balance between missed deserving cases and false promotions, probability calibration should remain part of operational use and be aligned with organizational costs.

On the other hand, the work contributes a methodological template for HR analytics. To our knowledge, and barring oversight, studies on promotion and related tasks such as turnover or recruitment rarely combine explicit equations, named function calls, Holm-adjusted significance tests, and a systematic reading of confusion matrices within a single, leakage-aware pipeline. By assembling these elements, the paper serves both researchers and practitioners as a practical reference they can replicate and adapt.

Nevertheless, generalizability should be probed through external validation on different organizations and time periods, alongside monitoring for calibration and model drift. Future work could explore transformer-based or hybrid approaches that scale to larger and more heterogeneous data. Deep learning architectures such as LSTM and CNN may

uncover temporal or high-dimensional patterns that are hard to capture with manual features. Alternative imbalance remedies, including ADASYN and Borderline-SMOTE, are worth testing in conjunction with threshold calibration. Broadening covariates while auditing fairness and subgroup robustness would improve inclusion and trust.

Ethical considerations

Not applicable.

Competing Interests

The authors confirm that there are no conflicts of interest.

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